WEEK 4

Mobile Application Testing Lab

Name: Vijayashree B
Domain: Cybersecurity (VAPT)
Date of Submission: 03.04.26

Mobile Application Testing Lab

1. Aim

To analyze the Android application InsecureBankv2 for security vulnerabilities using static and dynamic testing techniques and identify issues such as insecure storage, weak authentication, and exposed components.

2. Objectives

- To perform static analysis of the application using MobSF
- To understand dynamic testing concepts using Frida
- To analyze application components using Drozer
- To identify vulnerabilities such as insecure storage and authentication flaws
- To document findings and suggest security improvements

3. Tools Used

- MobSF
  Used for static analysis of the Android APK.
- Frida (Conceptual)
  Used for runtime manipulation and authentication bypass.
- Drozer
  Used to analyze inter-process communication and exported components.

4. Target Application

The application analyzed in this experiment is:

👉 InsecureBankv2

It is a deliberately vulnerable Android banking application designed for security testing and learning purposes.

5. Methodology

1. Static Analysis using MobSF

Static analysis was performed without executing the application.

Steps Performed:

1. MobSF server was started in Kali Linux
2. The APK file of InsecureBankv2 was uploaded
3. The tool performed automated analysis
4. Various sections such as permissions, code analysis, and storage were examined

Observations:

- Hardcoded credentials were found in the application
- Sensitive data was stored in plaintext
- SharedPreferences were used without encryption
- Local storage mechanisms were insecure

6. Identified Vulnerability

Test ID	Vulnerability	Severity	Target App
016	Insecure Storage	High	InsecureBankv2.apk

7. Description

The application stores sensitive data such as usernames, passwords, and banking details in plaintext within local storage and source code. This lack of encryption allows attackers to easily retrieve confidential information through reverse engineering or unauthorized access.

8. Impact

- Exposure of user credentials
- Unauthorized access to banking data
- Risk of financial fraud
- Compromise of sensitive user information

9. Recommendation

- Use encrypted SharedPreferences
- Implement Android Keystore system
- Avoid hardcoding sensitive data
- Follow secure coding practices

10. 2. Dynamic Testing (Frida Concept)

Dynamic testing was conceptually performed to analyze runtime behavior.

11. Analysis:

- Authentication functions were targeted
- Login validation logic was analyzed
- Client-side authentication was identified
- Authentication bypass was possible

12. Summary (50 words):

Frida was used to dynamically analyze InsecureBankv2 by hooking authentication functions at runtime. The login validation logic was overridden to bypass credential verification. This allowed unauthorized access without valid credentials, demonstrating weak client-side security and highlighting the need for secure server-side authentication mechanisms.

13. Observation:

The application relies heavily on client-side authentication, making it vulnerable to bypass attacks.

14. 3. IPC Testing using Drozer

Drozer was conceptually used to analyze the application's components and communication.

15. Findings:

- Several activities were exported
- Some components lacked proper permission checks
- Content providers were accessible externally
- Potential unauthorized access points were identified

16. Analysis:

Drozer was used to analyze the attack surface of InsecureBankv2. It revealed that certain application components such as activities and content providers were exported without proper access control, allowing external interaction and increasing the risk of unauthorized data access.

17. Checklist

- ✓ Run MobSF for static analysis
- ✓ Upload APK and review vulnerabilities
- ✓ Identify insecure storage

- ✓ Perform dynamic testing (Frida concept)
- ✓ Analyze IPC using Drozer
- ✓ Document findings

18. Results

The application InsecureBankv2 was successfully analyzed using static and conceptual dynamic techniques. The analysis revealed critical vulnerabilities including insecure storage of sensitive data, weak authentication mechanisms, and exposed application components.

19. Discussion

The experiment highlights common security issues in mobile applications. InsecureBankv2 demonstrates how improper handling of sensitive data and reliance on client-side authentication can lead to serious vulnerabilities.
Static analysis revealed insecure storage practices, while dynamic analysis showed authentication weaknesses. IPC testing further exposed risks associated with improperly secured components.
These findings emphasize the importance of implementing secure storage, proper authentication, and strict access controls in mobile applications.

20. Conclusion

The mobile application security testing lab was successfully completed using MobSF, Frida, and Drozer. The application InsecureBankv2 was found to contain critical vulnerabilities such as insecure storage, weak authentication, and exposed components.
Proper security mechanisms such as encryption, server-side validation, and secure component configuration must be implemented to mitigate these risks.